Lead Scoring Assignment - Summary Report

Objective of the Assignment -

X Education is a company that teaches people online. They sell classes to people who already work in certain fields. This company gets a lot of people in their classes every day. These people come from their website, search engines, social media, friends and sign up forms.. The problem is that not many of these people actually buy their classes.

This means the sales team wastes a lot of time talking to people who're not really interested. The goal of this project was to make a tool that can guess which people are most likely to buy classes from X Education. This tool gives each person a score from 0 to 100. If someone gets a score that means they are very interested in the classes. If they get a score that means they are not very interested. This will help the sales team focus on the people who're most likely to buy classes and make their job more efficient. X Education wants to use this tool to improve the way they sell their classes. The lead score, from the regression model will help them do that.

Data Understanding and Cleaning -

The dataset had 9240 rows and 37 columns. The target variable was Converted. This is where 1 means the lead was converted and 0 means the lead was not converted. The initial conversion rate was 38.54 percent. This shows that there was a lot of room for improvement.

When we cleaned the data we treated the value Select as a value. This is because it did not really mean anything in terms of what the user selected. We got rid of columns that had a lot of missing values. This is because they could make the model less reliable. We also dropped identifier columns like Prospect ID and Lead Number. This is because they do not really help us predict if a lead will be converted. We removed columns that did not have a lot of variation. This is because they did not add value to our predictions. We handled the remaining missing values in a way that made sense. For example we used mode imputation for variables and median imputation, for numerical variables. The dataset still had Converted as the target variable. We wanted to improve the conversion rate of the leads.

Exploratory Data Analysis-

We looked at the data to find patterns that could help us understand how people convert. What we found out was that people who used the Lead Add Form were very likely to convert. We also saw that people who came from the Welingak Website and from References were more likely to convert. It was also clear that people who are Working Professionals were more likely to convert which makes sense because X Education is mainly for people who work in industries.

When we looked at the numbers we saw that people who converted spent time on our website than people who did not convert. This tells us that how much people engage with our website is a sign of whether they will buy something or not. We also looked at things like what people did last on our website and what their last notable activity was. What we found was that things, like sending an SMS were associated with people who were more likely to convert. The X Education data showed us that conversion is related to how people use the website and what they do on it like Lead Add Form and other things.

Model Building Approach -

After cleaning and exploring the data I converted the variables into dummy variables. This was necessary so that I could use them in the regression model.

The dataset was then split into two parts: a training set and a testing set. I used a 70:30 ratio for this split.

- I scaled the variables, which included TotalVisits Total Time Spent on Website and Page Views Per Visit. I used StandardScaler for this purpose.
- I chose regression for this project because the target variable is binary.
- To select the features I used several methods.
- I used Recursive Feature Elimination (RFE) to identify the important variables.
- I checked the p-values to remove variables that were not statistically significant.
- I used Variance Inflation Factor (VIF) to check for multicollinearity.
- The final model had p-values that were acceptable.
- Also all VIF values were below 5.
- This meant that the model was statistically sound and did not have multicollinearity issues.
- The model was ready, for evaluation.

Model Evaluation -

The final model did a job on the training data and the test data. It got 80.1 percent of the training data correct and about 80.2 percent of the test data correct. The model was pretty good at finding the leads that would actually work out. It found 83.8 percent of them in the training data and about 83.7 percent of them in the test data.

The model was also okay at figuring out which leads would not work out. It got 77.7 percent of them right in the training data and about 77.9 percent of them right in the test data.

The model scored 0.88 on the AUC test for both the training data and the test data, which means the model is good at telling the difference between leads that will work out and leads that will not work out. The model uses a cutoff of 0.3 because it is a balance, between finding leads that will work out and avoiding leads that will not work out.

Lead Scoring and Business Recommendations -

The predicted chances were turned into lead scores by multiplying them by 100. Leads with scores are hot leads and should be contacted first.
Based on our model the top positive predictors are:
Lead Origin. Lead Add Form, What is your current occupation. Working Professional, Lead Source. Welingak Website X Education should focus on leads from Lead Add Form, Welingak Website and working professionals.

Leads who spend time on the website show higher engagement and should be contacted quickly.

Here are some tips, for sales periods:

- During sales periods use a lower cutoff to get more potential leads.
- When the target is already achieved use a cutoff to avoid unnecessary calls.

Learnings from the Assignment -

This assignment helped in understanding how logistic regression can be applied to a real business problem. It also highlighted the importance of data cleaning, missing value treatment, EDA, feature selection, model evaluation, and converting model results into business recommendations. The final model can help X Education improve sales productivity by allowing the sales team to focus on the most promising leads.